PEEYOOSH KANGLE

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SUMMARY

Motivated Computer Science Engineer with a strong academic foundation (**Cumulative CGPA: 8.44, Sem VI: 9.22**) and practical expertise in building scalable Relational Database Architectures, ETL Data Ingestion Pipelines, and Backend Microservices using **SQL, Python, C++, and Node.js**. Skilled in 3NF schema design, complex query optimization, indexing strategies, and transactional integrity (ACID). Experienced in RESTful API engineering, automated CI/CD testing, and solving 150+ algorithmic problems. Seeking a **Software Development Engineer (SDE) / Backend Developer** role to architect robust, high-performance enterprise solutions.

TECHNICAL COMPETENCIES

Languages:	Python, SQL, JavaScript (ES6+), C++
Databases & RDBMS:	MS SQL Server, PostgreSQL, MySQL, SQLite, Oracle (Fundamentals)
Core DB Concepts:	Relational Schema Design, Normalization (1NF–3NF), ACID Properties, Complex Joins, Indexing, Query Optimization
Backend Frameworks:	Node.js, Express.js, FastAPI, RESTful APIs, CRUD Workflows, Postman, WebSockets
Data & Analytics:	Pandas, NumPy, Scikit-learn, Data Pipelines, Exploratory Data Analysis (EDA), Automated Reporting
DevOps & Tools:	Git, GitHub, Docker (Basics), Linux/Bash Scripting, VS Code, CI/CD Fundamentals, SDLC
AI & Retrieval:	Retrieval-Augmented Generation (Local RAG), Vector Search, Sentence-Transformers, Ollama API

WORK EXPERIENCE & PROJECTS

Intelligent Student Management & Academics Portal — GitHub ↗ — Jan 2026 — Present

Tech Stack: Node.js, PostgreSQL, Express.js, SQL, REST APIs, Postman

- Architected an enterprise **3NF-normalized relational schema** across 12+ tables with foreign keys and strict **ACID transactional integrity**.
- Engineered optimized SQL queries, subqueries, and multi-table joins, **reducing query latency by 45%** through B-tree index tuning.
- Built and validated **20+ secure RESTful API endpoints** using Postman, ensuring robust input sanitization and structured error logging.
- Automated attendance and CGPA calculation workflows, eliminating **90% of manual administrative data entry**.

Jarvis AI — Multimodal Voice Assistant & Knowledge Retrieval — GitHub ↗ — Sep 2025 — Dec 2025

Tech Stack: Python, Local RAG, Sentence-Transformers, Vector Search, Ollama API, Multi-Threading

- Engineered an on-premise **Retrieval-Augmented Generation (RAG)** pipeline querying vector embeddings, delivering **sub-second semantic search** across unstructured documentation.
- Implemented multi-threaded concurrency for audio ingestion and vision (DeepFace), **improving interface responsiveness by 55%** without UI thread deadlock.
- Integrated local RESTful Ollama endpoints with persistent session caching to retain conversational context across multi-turn dialogues.

SmartCrop — Precision Farming & Predictive Analytics Pipeline — GitHub ↗ — Jun 2025 — Aug 2025

Tech Stack: Python, Scikit-learn, Pandas, NumPy, Streamlit, Data Validation

- Constructed end-to-end data processing pipelines ingesting **2,200+ agricultural records**, automating null imputation, outlier rejection, and feature scaling.
- Conducted exploratory data analysis (EDA) and trained comparative ML models (Random Forest vs. SVM), achieving an **industry-grade 99.31% model accuracy**.
- Built an interactive analytics dashboard featuring real-time parameter simulation and comparative accuracy reports.

CyberShield — AI Cybercrime & Investigation Management Platform — GitHub ↗ — Live Demo ↗ — Jan 2026 — Jun 2026

Tech Stack: React 19, Node.js, Express.js, MongoDB, JWT, RBAC, Helmet, SHA-256 Hashing

- Engineered a centralized cybercrime investigation management system featuring **Role-Based Access Control (Citizen, Investigator, Admin)** and automated tracking IDs (`CS-2026-XXXXXX`).
- Implemented cryptographic digital evidence hashing complying with **Indian Evidence Act (Section 65B)** guidelines, automated PDF incident reports, and real-time case tracking.
- Secured REST API endpoints using **JWT authentication**, **bcrypt password hashing (12 salt rounds)**, Helmet HTTP security headers, and rate limiting against brute-force attacks.
- **Selected for the open-source XERJ Corpus Hub (Austin, TX)** as a verified reference architecture for AI coding agents and developer search tools.

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science & Engineering — 2023 — 2027

Jawaharlal Nehru Engineering College (JNEC), MGM University, Chhatrapati Sambhajinagar — **Cumulative CGPA: 8.44**

Semester Grades (SGPA): Sem 1: **7.99** | Sem 2: **8.36** | Sem 3: **7.97** | Sem 4: **8.54** | Sem 5: **8.58** | Sem 6: **9.22**

- Higher Secondary Certificate (Class XII): Rajarshi Shahu Mahavidyalaya, Latur | Science | Percentage: **71.00%**
- Secondary School Certificate (Class X): Kholeshwar High School, Ambajogai | Percentage: **99.60%**

CERTIFICATIONS & ACHIEVEMENTS

- **150+ Algorithmic Problems Solved (LeetCode Top 150)** — Implemented in modern C++17 and Python 3.12 (DP, Graph Traversal, Trees, Sliding Window) with automated GitHub Actions CI regression suites.
- **SQL (Intermediate) Certified** — HackerRank (Verified in Complex Joins, Subqueries, Aggregations, and Indexing).
- **National Cadet Corps (NCC):** Certified NCC Cadet; holder of prestigious **NCC 'A', 'B', and 'C' Certificates**; attended Indian Army Training Camp; Blood Donation volunteer.
- **Open-Source Recognition — XERJ Corpus Hub:** CyberShield architecture selected by the founder of XERJ (Austin, TX) as an official reference implementation for AI coding agents.
- **The Ultimate Job Ready Data Science Course** — CodeWithHarry (Comprehensive Python, Pandas, EDA, and Machine Learning).